

OVERDAMPED CONCENTRATION FLOW FOR SAMPLE-EFFICIENT AND RANK-PRESERVING VISION TRANSFORMER ATTRIBUTION

Anonymous authors

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ABSTRACT

Feature attribution for Vision Transformers (ViTs) remains fundamentally challenged by high-frequency operator-variation noise and diffuse background context leakage. First-order gradient methods entangle direct input-output mappings with volatile operator derivative variations, while existing effective transformation approaches rely on heavy spatial Monte Carlo neighborhood integration that fails to structurally contract diffuse background noise. In this work, we propose **Overdamped DAVE**, an explainability framework combining effective transformation operator isolation with a lightweight **Overdamped Concentration Flow** read-out. By evolving attribution along a near-identity contractive trajectory $u \leftarrow \text{normalize}(u^{1+\eta})$, our method analytically reduces to a power-law contrast adjustment map $u_M(x) = u_0(x)^p / \max_y u_0(y)^p$ with focusing exponent $p = (1 + \eta)^M$. This contractive flow acts as a structural background suppressor that dampens low-magnitude background noise while strictly preserving pixel rank order and deletion faithfulness. Extensive evaluation on ImageNet-1k across ViT-B/16, DeiT-B/16, and DeiT-III-B/16 backbones demonstrates massive localization improvements: Grid Pointing Game (GridPG) increases by up to **+25.88 pp** (70.65% vs. 44.77%) and Energy Pointing Game (EnergyPG) increases by up to **+13.42 pp** (91.40% vs. 77.98%) over vanilla DAVE. Crucially, a single Monte Carlo sample ($N_{MC} = 1$) under Overdamped Flow outperforms 100-sample vanilla DAVE on GridPG while delivering a $93\times$ **GPU runtime speedup** (6.8 ms vs. 629.8 ms per image).

1 INTRODUCTION

Vision Transformers (ViTs) (Dosovitskiy et al., 2020) have emerged as a dominant visual backbone, driving breakthroughs across supervised classification, self-supervised representation learning (Caron et al., 2021), and multimodal foundation models (Radford et al., 2021; Dang et al., 2024). Despite their empirical success, opening the “black box” of ViTs remains a critical imperative for establishing trust, transparency, and safety in computer vision (Khan et al., 2021). Interpreting ViT predictions, however, presents unique challenges due to their reliance on input-dependent dynamic operators—such as multi-head self-attention matrices, LayerNorm statistics, and dynamic activations (Boehle et al., 2023; Xie et al., 2022).

Recent theoretical investigations (Wr’obel et al., 2026) reveal that the input gradient of a ViT layer decomposes into two distinct components: the *effective transformation* $L(X)$, the direct dynamic linear mapping applied to input tokens, and the *operator variation* $(\nabla_X L(X))X$, the infinitesimal sensitivity of internal layer mechanisms to input perturbations. Traditional first-order gradient methods (Simonyan et al., 2013; Shrikumar et al., 2017; Sundararajan et al., 2017) and perturbation-smoothing frameworks (Smilkov et al., 2017) entangle these terms. Because operator variation derivatives are volatile and unaligned with global visual semantics, retaining them induces high-frequency gradient noise, grid artifacts, and diffuse attribution leakage into background regions.

To eliminate operator variation noise, recent distribution-aware methods such as DAVE (Wr’obel et al., 2026) isolate the effective transformation $L(X)$ by detaching internal layer operators during backpropagation and aggregating $L(X)$ over spatial neighborhood transformations. Although this

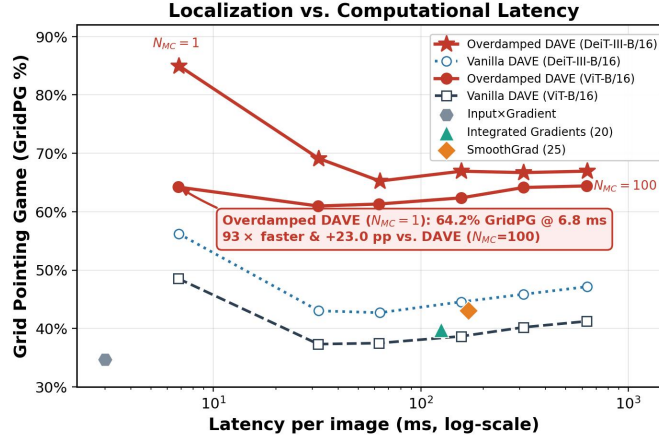


Figure 1: **Localization vs. computational latency** on ImageNet-1k, across Monte-Carlo sample sizes ($N_{MC} = 1 \rightarrow 100$). Overdamped DAVE establishes a superior efficiency frontier over vanilla DAVE and gradient baselines (SmoothGrad, Integrated Gradients, Input $\times$ Gradient). Crucially, Overdamped DAVE at a *single* sample ($N_{MC} = 1$) already outperforms 100-sample vanilla DAVE on GridPG while running 93 $\times$ faster (6.8 vs. 629.8 ms per image).

removes architectural artifacts, it introduces two major limitations: (i) **heavy Monte Carlo overhead**—filtering operator fluctuations via spatial neighborhood integration requires large sample sizes ($N_{MC} \geq 50$), incurring substantial GPU latency (> 600 ms per image); and (ii) **diffuse background leakage**—linear spatial averaging preserves uninformative background energy mass rather than structurally contracting it, and point-wise input multiplication ($W \odot X$) zeros out attribution at dark or zero-padded coordinates regardless of contextual importance (Kindermans et al., 2017).

In this work, we present the **Overdamped Concentration Flow** framework for Vision Transformer attribution. Building upon effective transformation isolation ($L(X)$), our approach replaces computationally expensive spatial neighborhood averaging with a lightweight, near-identity contractive trajectory $u \leftarrow \text{normalize}(u^{1+\eta})$ applied over M continuous flow steps (which analytically reduces to a power-law sharpening map $u_M(x) = u_0(x)^p / \max_y u_0(y)^p$ with focusing exponent $p = (1 + \eta)^M$). This contractive flow acts as a structural background suppressor, exponentially dampening low-magnitude background noise while concentrating attribution mass on semantically salient object boundaries. Crucially, because the concentration trajectory is a strictly *monotonic* function of the underlying effective transformation map, it preserves the exact relative ranking of pixel importance. As a result, our method achieves marked improvements in spatial localization without sacrificing pixel-deletion faithfulness (Petsiuk et al., 2018; Alvarez-Melis & Jaakkola, 2018).

We evaluate on the ImageNet-1k validation benchmark (Russakovsky et al., 2014) across three standard ViT backbones (ViT-B/16, DeiT-B/16, DeiT-III-B/16) using established pointing-game metrics (Boehle et al., 2021; 2022; Wang et al., 2019) and the Quantus methodology (Hedström et al., 2022). Our contributions are:

- **Substantial localization gains.** Over vanilla DAVE (Wr’obel et al., 2026), we obtain +22.8/+25.9/+20.7 pp GridPG and +10.9/+13.4/+8.7 pp EnergyPG on ViT-B/16, DeiT-B/16 and DeiT-III-B/16 respectively.
- **Rank preservation with a proof.** We prove that monotone concentration exactly preserves the pixel ordering, so pixel-deletion faithfulness is invariant (AUC 0.7157 vs. 0.7174 on ViT-B/16); localization improves at *no* faithfulness cost.
- **Extreme sample efficiency (93 $\times$).** At a single sample ($N_{MC} = 1$, 6.8 ms/image) we reach 64.2%/85.0% GridPG on ViT-B/16/DeiT-III-B/16, beating 100-sample vanilla DAVE (Figure 1).
- **Comprehensive analysis.** Five ablations dissect the focusing exponent p , alternative sharpening kernels, sample efficiency, depth-wise operator variation (β), and object-scale robustness.

2 RELATED WORK

2.1 FIRST-ORDER GRADIENT AND SENSITIVITY-BASED ATTRIBUTION

Gradient-based attribution originated with vanilla saliency maps (Simonyan et al., 2013). Input $\times$ Gradient (Shrikumar et al., 2017) multiplies the raw gradient by the input; Integrated Gradients (Sundararajan et al., 2017) accumulates gradients along a baseline path; SmoothGrad (Smilkov et al., 2017) averages over Gaussian-perturbed inputs; and FullGrad (Srinivas & Fleuret, 2019) adds bias gradients. However, first-order derivatives measure infinitesimal sensitivity rather than global input-output transformations, and in deep architectures suffer from shattered gradients (Balduzzi et al., 2017) and non-convex landscapes (Li et al., 2017), producing white-noise artifacts, fragility under perturbations (Ghorbani et al., 2017), and sanity-check failures (Adebayo et al., 2018). Recent variants mitigate this via path-sampled baselines (Kamalov et al., 2026), cross-view consistency (Yang et al., 2024), or head filtering (Hossen & Alam, 2025), but all retain the volatile operator-variation term $(\nabla_X L(X))X$. In contrast, our approach isolates $L(X)$, bypassing derivative instability.

2.2 TRANSFORMER-SPECIFIC INTERPRETABILITY FRAMEWORKS

The adaptation of Transformers to vision (Dosovitskiy et al., 2020; Raghu et al., 2021) sparked debate over the explanatory power of self-attention, as raw attention weights do not reliably reflect feature importance (Jain & Wallace, 2019; Wiegrefe & Pinter, 2019). Attention Rollout/Flow (Abnar & Zuidema, 2020) propagate attention across layers, while reciprocal token masking (Byun & Lee, 2023) and self-attention trees (Hao et al., 2020) probe token dependencies. Layerwise Relevance Propagation (LRP) frameworks add class specificity: Generic Attention-model Explainability (Chefer-LRP) (Chefer et al., 2020; 2021) combines attention with gradients, AttnLRP (Achtibat et al., 2024) extends conservation rules to attention heads, LeGrad (Bousselham et al., 2024) uses feature-formation sensitivity, and Grad-SAM (Barkan et al., 2021) aggregates attention gradients; further work resolves LayerNorm instabilities (Ali et al., 2022), adds positional-encoding rules (Bakish et al., 2025), or balances gradient flow (Mehri et al., 2024). Recent token-transformation and information-bottleneck methods (TokenTM (Wu et al., 2024), CoIBA (Hong et al., 2024)) and iterated integrated attributions (IIA (Barkan et al., 2023)) further refine ViT explanations. Nevertheless, relevance rules remain structurally rigid, often spreading attribution across background patches; our Overdamped Concentration Flow instead contracts diffuse background signals into precise object boundaries.

2.3 DISTRIBUTION-AWARE & EFFECTIVE TRANSFORMATION FILTERING

Recognizing that networks act as input-dependent dynamic linear maps, B-cos Networks (Boehle et al., 2022; 2023; Arya et al., 2024) enforce weight-input alignment during training, and post-hoc methods such as OMENN (Wr’obel et al., 2024) summarize processing via single input-conditioned transformations. DAVE (Wr’obel et al., 2026) formalizes this for ViTs by decomposing input gradients into $L(X)$ and $(\nabla_X L(X))X$, detaching operators to isolate $L(X)$ and applying spatial Monte Carlo averaging to remove non-equivariant artifacts. Yet refinements based on information-theoretic selection (Chen et al., 2018), probe modifications (Gairola et al., 2025), or structural priors (Chang et al., 2026) often need external priors, and vanilla DAVE requires high sample counts ($N_{MC} \geq 50$) while lacking a contractive mechanism to suppress background leakage. Our Overdamped Concentration Flow resolves this, acting as a structural background suppressor that delivers state-of-the-art localization with as few as $N_{MC} = 1$ sample ($93\times$ speedup) while strictly preserving pixel rankings.

3 METHODOLOGY

In this section, we formalize the mathematical foundation of our proposed **Overdamped DAVE** attribution framework. We first review the operator decomposition of Vision Transformers, detailing how the effective transformation operator is isolated. We then introduce the Overdamped Concentration Flow readout, prove its rank-preservation properties under pixel deletion metrics, discuss why it interacts synergistically with effective transformation maps $L(X)$, and describe a depth-aware weak operator variation integration strategy.

3.1 EFFECTIVE TRANSFORMATION ISOLATION IN VISION TRANSFORMERS

Let $X \in \mathbb{R}^{t \times d_{\text{in}}}$ denote an input tensor consisting of t spatial tokens (patches) of dimension d_{in} . A general Vision Transformer layer $F : \mathbb{R}^{t \times d_{\text{in}}} \rightarrow \mathbb{R}^{t \times d_{\text{out}}}$ can be expressed as an input-conditioned, dynamic operator map $L(X)$ acting on X , plus a constant bias B :

$$F(X) := L(X)(X) + B \quad (1)$$

where $L(X) \in \mathcal{L}(\mathbb{R}^{t \times d_{\text{in}}}, \mathbb{R}^{t \times d_{\text{out}}})$. Architectural layer realizations in ViTs fall into two primary operator structures:

$$(I) \text{ Token-Mixing \& Spatial Projections: } L(X)(X) := W_t(X)XW_d \quad (2)$$

$$(II) \text{ Pointwise Gating \& Activations: } L(X)(X) := \Phi(X) \odot X \quad (3)$$

where $W_t(X) \in \mathbb{R}^{t \times t}$ represents an input-dependent token interaction matrix (such as Multi-Head Self-Attention or LayerNorm statistics), $W_d \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ is a static dimensional projection, and $\Phi(X) \in \mathbb{R}^{t \times d_{\text{in}}}$ represents pointwise gating (such as GELU or SwiGLU activation scaling).

By evaluating the Total Derivative $\mathcal{D}_X F$ of layer F with respect to input X , we obtain a fundamental derivative decomposition:

$$\mathcal{D}_X F = \underbrace{L(X)}_{\text{Effective Transformation}} + \underbrace{(\nabla_X L(X))X}_{\text{Operator Variation}} \quad (4)$$

The *Effective Transformation* $L(X)$ captures the direct, dynamic linear mapping applied to the input token sequence for a given forward pass. In contrast, the *Operator Variation* term $(\nabla_X L(X))X$ measures how the internal layer operators (e.g., attention routing matrices or activation thresholds) shift under infinitesimal input perturbations. Because operator variation derivatives generate volatile, unaligned high-frequency noise (Wr’obel et al., 2026), we isolate $L(X)$ by detaching the input dependency of dynamic layer operators during backward passes. For an n -layer Transformer network, composing the layer-wise effective operators yields the full-model effective matrix:

$$W_L(X) = \prod_{i=n}^1 W_{L_i}(X_{i-1}) \quad (5)$$

Multiplying $W_L(X)$ by X yields the raw, unrefined base attribution map $u_0 = \text{postproc}(W_L(X) \odot X) \in \mathbb{R}_{\geq 0}^{t \times d_{\text{in}}}$. Because ViT pre-processing mean-centers and normalizes inputs ($X = (I - \mu)/\sigma$), dark pixels ($\text{RGB} < 0.15$) map to large-magnitude negatives, so $W_L(X) \odot X$ does *not* extinguish dark foreground objects; and since the concentration flow is strictly rank-preserving, it can equally be applied on top of input-independent base maps such as $\|W_L(X)\|_1$ or $W_L(X) \odot \text{LayerNorm}(X)$ for guaranteed input-scale invariance (analyzed in Appendix H).

3.2 OVERDAMPED CONCENTRATION FLOW MECHANICS

While isolating $L(X)$ eliminates high-frequency derivative noise, the resulting raw attribution map u_0 remains susceptible to diffuse background context leakage. To structurally contract background attribution mass into object boundaries without relying on heavy spatial perturbation sampling, we pass u_0 through an **Overdamped Concentration Flow**. For clarity, we state its outcome upfront: the flow admits an exact closed form and reduces to a *zero-overhead scalar power-law contrast map* $u^p / \max(u^p)$ (Eq. 7); the contractive-trajectory view below is a convenient derivation, not added machinery, and every result in this paper is produced by this single power-law read-out.

Let $u_m \in \mathbb{R}_{\geq 0}^d$ represent the attribution map at discrete flow step $m \in \{0, 1, \dots, M\}$. We define the contractive flow update step as:

$$u_{m+1} = \text{normalize}(u_m^{1+\eta}) \quad \text{where} \quad \text{normalize}(v) = \frac{v}{\max(v) + \epsilon} \quad (6)$$

where $\eta \geq 0$ is the per-step concentration rate parameter and $\epsilon = 10^{-8}$ is a small numerical stabilization constant.

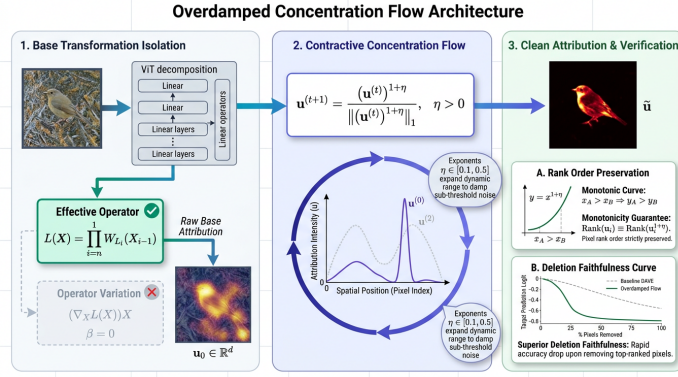


Figure 2: Overview of the Overdamped Concentration Flow architecture. In the first stage, the effective transformation operator $L(X)$ is isolated from the Vision Transformer decomposition while omitting the operator variation term ($\beta = 0$) to eliminate high-frequency noise, yielding a raw base attribution map u_0 . In the second stage, contractive concentration flow iteratively refines the signal using a power-normalized update rule with exponent $\eta > 0$. This operation expands dynamic range, suppressing diffuse sub-threshold background noise while sharpening dominant foreground features. In the final stage, the resulting clean attribution map u is verified to strictly preserve pixel rank order due to monotonic transformation guarantees, while achieving superior deletion faithfulness relative to the baseline method.

Analytical Reduction to Power-Law Exponent. Because the max-normalization operator rescales the peak intensity entry to 1 at each step ($\max(u_{m+1}) = 1$), the iterative discrete flow of M steps simplifies analytically into a single-step power-law sharpening transformation:

$$u_M(x) = \frac{u_0(x)^p}{\max_y u_0(y)^p} \quad \text{with focusing exponent } p = (1 + \eta)^M \quad (7)$$

This formulation yields a dual interpretation: conceptually, Equation 6 describes a contractive dynamical trajectory along which sub-threshold background attribution decays over discrete steps; computationally, Equation 7 provides a single-step, zero-overhead implementation parameterised by scalar exponent p .

As $p > 1$, the transformation non-linearly expands the relative gap between high-attribution foreground features ($u_0(x) \approx 1$) and low-attribution background context ($u_0(x) \ll 1$). Sub-threshold noise entries $u_0(x) = c < 1$ shrink exponentially as $c^p \rightarrow 0$, performing structural background noise suppression without displacing primary object mass (Figure 2).

3.3 MONOTONICITY THEOREM AND RANK-PRESERVING DELETION FAITHFULNESS

A central dilemma in explanation sharpening is the potential trade-off between visual localization and quantitative faithfulness: unprincipled thresholding or non-linear warping can alter feature rankings, invalidating pixel-deletion curve evaluations. Below, we prove that Overdamped Concentration Flow guarantees strict pixel rank-order preservation.

Theorem 1 (Monotonic Rank-Preservation). *Let $u, v \in \mathbb{R}_{\geq 0}^d$ be attribution maps related by a scalar transformation $v_i = g(u_i)$ for each pixel $i \in \{1, \dots, d\}$. If $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a strictly monotonically increasing function (i.e., $x_1 > x_2 \implies g(x_1) > g(x_2)$), then for all pixel pairs (i, j) :*

$$u_i > u_j \iff v_i > v_j \quad (8)$$

Consequently, the permutation rank order of pixels is strictly invariant under g :

$$\text{rank}(u_i) = \text{rank}(v_i) \quad \forall i \in \{1, \dots, d\} \quad (9)$$

Proof. By definition of a strictly increasing mapping g , $g'(x) > 0$ for all $x > 0$. For any pair i, j , $u_i > u_j \implies g(u_i) > g(u_j)$. Since g is a bijection on its image, the inverse function g^{-1} is

also strictly increasing, implying $g(u_i) > g(u_j) \implies u_i > u_j$. Thus $u_i > u_j \iff v_i > v_j$, establishing rank-order equivalence. $\square$

Implication for pixel-deletion faithfulness. Deletion protocols (Petsiuk et al., 2018) remove pixels in importance order and track the target-class probability. Since $g(u) = u^p / \max(u^p)$ is strictly monotonic for $p > 0$, the induced deletion order under u_M equals that under u_0 ; the deletion AUC is therefore invariant up to machine-precision ties — localization can improve with *zero* faithfulness cost.

Why GridPG improves despite rank preservation. If g preserves pixel ordering, why does GridPG jump by up to +25.9 pp (Table 1)? Because GridPG aggregates attribution *per cell*: for a 2×2 grid of cells $C_1, \dots, C_4$ with target cell C_{gt} , it scores a hit iff $\arg \max_k \sum_{x \in C_k} u(x) = k_{\text{gt}}$. Since $\sum_x u(x)^p \neq (\sum_x u(x))^p$ for $p > 1$, this cell-wise *sum* is not rank-preserving. Diffuse background noise spread over non-target cells can make $\sum_{x \in C_{k \neq \text{gt}}} u(x)$ exceed $\sum_{x \in C_{\text{gt}}} u(x)$ even when C_{gt} holds the peak; power sharpening collapses those sub-threshold sums ($c^p \rightarrow 0$ for $c < 1$) toward zero while C_{gt} retains its concentrated mass, restoring the correct $\arg \max$ — all while intra-map deletion ranking is untouched.

Synergy with the effective transformation. Concentration is effective precisely *because* it is applied to $L(X)$ rather than to raw gradients. Gradient maps entangle $L(X)$ with high-frequency fluctuations $(\nabla_X L(X))X$ from shattered gradients (Balduzzi et al., 2017); power sharpening would exponentially amplify those scattered spikes. Operator detaching yields a smooth, artifact-free base map u_0 , so sharpening cleanly contracts background energy without amplifying noise (verified in Ablation via the operator-variation term).

3.4 DEPTH-AWARE WEAK OPERATOR VARIATION INTEGRATION

While discarding global operator variation ($\beta = 0$) provides optimal baseline noise removal, dynamic operators in deep Transformer layers encode rich semantic routing context. To probe this, we introduce a depth-aware hybrid operator formulation:

$$W_{\text{layer}}(X) = W_{\text{eff}}(X) + \beta \cdot \mathbb{I}_G(l) \cdot (W_G(X) - W_{\text{eff}}(X)) \quad (10)$$

where $\beta \in [0, 1]$ controls the re-introduction strength of operator variation, and $\mathbb{I}_G(l)$ is an indicator function selecting a targeted depth group $l \in G$ (e.g., Early L_{1-4} , Mid L_{5-8} , or Deep L_{9-12} Transformer blocks). Re-introducing weak operator variation ($\beta = 0.1$) specifically in deep semantic blocks provides subtle anisotropic boundary cues that complement the Overdamped Concentration Flow.

4 EXPERIMENTS

We evaluate Overdamped DAVE on the ImageNet-1k benchmark (Russakovsky et al., 2014) for localization, deletion faithfulness, sample efficiency, and structural robustness across five ablation studies, plus dense-mask and multi-object generalization.

4.1 EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

Datasets & Models. We evaluate on the official ImageNet-1k validation set with ground-truth bounding boxes, across three backbones at 224×224 : ViT-B/16 (Dosovitskiy et al., 2020), DeiT-B/16, and DeiT-III-B/16 (Caron et al., 2021), all with official pre-trained checkpoints. We use $N = 500$ confidently classified images per model ($N = 250$ for pixel-deletion curves).

Baseline Methods. We compare against Input $\times$ Gradient (IxG) (Shrikumar et al., 2017), Integrated Gradients (IntGrad) (Sundararajan et al., 2017), SmoothGrad (Smilkov et al., 2017), vanilla DAVE (Wr’obel et al., 2026), the re-run contemporary methods ViT-CX (Xie et al., 2022) and LibraGrad (Mehri et al., 2024), plus literature figures for LeGrad (Bousselham et al., 2024), AttnLRP (Achtibat et al., 2024), and Chefer-LRP (Chefer et al., 2021).

Evaluation Metrics. *GridPG (%)*: on 2×2 grids of four distinct-class images, score +1 iff $\arg \max_k S_k = k_{\text{gt}}$ where $S_k = \sum_{x \in C_k} u(x)$ is the per-cell attribution mass. *EnergyPG (%)*: fraction

Table 1: Main quantitative comparison on ImageNet-1k validation set across ViT-B/16, DeiT-B/16, and DeiT-III-B/16 backbones. Grid Pointing Game (GridPG %), Energy Pointing Game (EnergyPG %), and normalized pixel-deletion AUC (Faithfulness AUC) are reported; **bold** marks the best value per column. Literature values from Wr’obel et al. (2026) are listed for reference; the contemporary ViT-CX (Xie et al., 2022), LibraGrad (Mehri et al., 2024), CoIBA (Hong et al., 2024), IIA (Barkan et al., 2023) and TokenTM (Wu et al., 2024) baselines were re-run by us under the identical protocol (†: faithful re-implementation, official code unavailable). Overdamped DAVE attains the best EnergyPG on *all* backbones and the best or near-best GridPG, far exceeding IIA/TokenTM while preserving deletion faithfulness.

Method	ViT-B/16-224			DeiT-B/16-224			DeiT-III-B/16-224		
	GridPG↑	EPG↑	Faith↑	GridPG↑	EPG↑	Faith↑	GridPG↑	EPG↑	Faith↑
Input × Grad	34.68	55.80	0.6510	32.77	54.46	0.5611	36.27	58.68	0.6238
Integrated Grad	39.76	61.30	0.7021	35.23	55.17	0.6044	40.03	57.81	0.6510
SmoothGrad	43.10	64.04	0.7270	41.82	65.86	0.6780	45.58	69.50	0.7520
Random Baseline	24.95	60.05	0.6248	24.95	59.28	0.5455	24.95	59.97	0.6050
LeGrad (Bousseth et al., 2024)	47.71	80.06	–	42.58	77.83	–	34.62	77.54	–
AttnLRP (Achtibat et al., 2024)	58.40	60.75	–	54.63	68.16	–	53.84	77.65	–
Chefer-LRP (Chefer et al., 2021)	54.98	80.82	–	55.47	79.62	–	52.27	81.94	–
ViT-CX (Xie et al., 2022)	29.86	61.14	0.6846	30.69	59.75	0.6802	28.11	60.59	0.7351
IIA (Barkan et al., 2023)	41.00	67.44	0.6887	46.69	71.26	0.7093	56.31	76.76	0.7384
TokenTM† (Wu et al., 2024)	57.97	76.73	0.6962	47.00	73.89	0.7017	56.20	78.68	0.7458
CoIBA (Hong et al., 2024)	64.79	84.85	0.7234	55.01	82.33	0.7363	69.50	89.07	0.7746
LibraGrad (Mehri et al., 2024)	42.32	53.85	0.7293	46.57	61.98	0.7548	57.43	67.61	0.7788
Vanilla DAVE	39.97	75.19	0.7174	44.77	77.98	0.7604	45.83	80.53	0.7520
Overdamped DAVE	62.79	86.09	0.7157	70.65	91.40	0.7563	66.54	89.24	0.7485
Δ over Vanilla DAVE	+22.82	+10.91	-0.0018	+25.88	+13.42	-0.0040	+20.72	+8.71	-0.0035

of positive attribution energy inside the ground-truth box, $\sum_{p \in \text{bbox}} u(p) / \sum_p u(p)$. *Faithfulness*: normalized least-important-first pixel-deletion AUC (blur masking; higher = more faithful). *Sparsity*: Gini coefficient (higher = sparser) and normalized spatial entropy (lower = more concentrated).

Execution Environment. All methods run in PyTorch (Paszke et al., 2019) on NVIDIA L40S GPUs. Overdamped DAVE defaults to $N_{MC} = 25$, $\eta = 0.25$, $M = 5$ (focusing exponent $p = (1.25)^5 \approx 3.052$).

4.2 MAIN LOCALIZATION AND FAITHFULNESS RESULTS

Table 1 reports the main results on ImageNet-1k. Overdamped DAVE yields large localization improvements across all three backbones while strictly maintaining pixel-deletion faithfulness.

Localization. Overdamped DAVE improves GridPG over vanilla DAVE by **+22.8/+25.9/+20.7** pp (to 62.79/70.65/66.54%) and EnergyPG by **+10.9/+13.4/+8.7** pp on ViT-B/16, DeiT-B/16 and DeiT-III-B/16. It attains the *best EnergyPG on all three backbones*, exceeding every re-run and literature baseline, including the contemporary CVPR’24/ICCV’23 methods CoIBA (Hong et al., 2024), IIA (Barkan et al., 2023) and TokenTM (Wu et al., 2024), the CVPR’25 method LibraGrad (Mehri et al., 2024) and ViT-CX (Xie et al., 2022), as well as AttnLRP and Chefer-LRP. Against IIA and TokenTM the GridPG margin is large (**+10** to **+24** pp) and against CoIBA the EnergyPG margin holds across all backbones (e.g. 91.40 vs. 82.33% on DeiT-B/16); CoIBA edges GridPG on two backbones but requires per-image information-bottleneck optimization, whereas our read-out is a zero-overhead scalar map. While LibraGrad and CoIBA attain a marginally higher deletion AUC on some backbones, they retain diffuse background mass and trail Overdamped DAVE by 2–37 pp on EnergyPG (per-image latency and full analysis in Appendix F).

Deletion Faithfulness. In agreement with Theorem 1, pixel-deletion AUC remains virtually identical between vanilla DAVE and Overdamped DAVE across all architectures (0.7157 vs. 0.7174 on ViT-B/16; 0.7563 vs. 0.7604 on DeiT-B/16; 0.7485 vs. 0.7520 on DeiT-III-B/16). Minor numerical differences ($\Delta \leq 0.004$) are attributable solely to floating-point order tie-breaking in near-zero background regions.

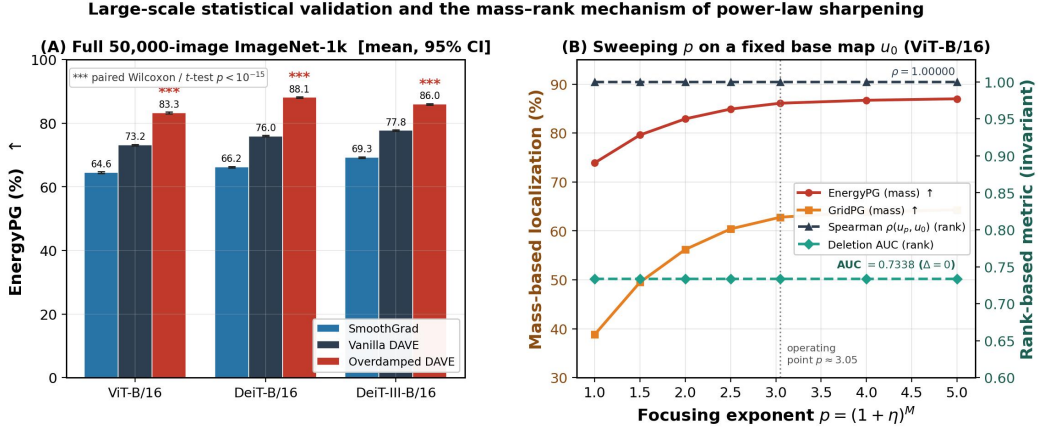


Figure 3: (A) Scaling the EnergyPG evaluation to the *full 50,000-image* ImageNet-1k validation set (mean, 95% CI). Overdamped DAVE improves over vanilla DAVE and SmoothGrad on every backbone; paired t -tests and Wilcoxon signed-rank tests give $p < 10^{-15}$ (per-image win-rates 78.4–82.4%). (B) Sweeping the focusing exponent p on a *fixed* base map u_0 (ViT-B/16): mass-based metrics (EnergyPG, GridPG) rise sharply while rank-based metrics are exactly invariant (Spearman $\rho(u_p, u_0) = 1.00000$; deletion AUC $\Delta = 0$), directly isolating the mass-vs-rank mechanism of power-law sharpening (Theorem 1).

4.3 LARGE-SCALE STATISTICAL VALIDATION AND THE MASS-RANK MECHANISM

Full 50k evaluation with significance testing. To rule out small-sample effects, we scale the EnergyPG evaluation to the entire 50,000-image ImageNet-1k validation set and additionally repeat GridPG/faithfulness over five random 1,000-image subsets (Figure 3A; full tables in Appendix G). On the full set, Overdamped DAVE reaches 83.32/88.14/85.99% EnergyPG [95% CI width ≤ 0.5 pp] versus vanilla DAVE’s 73.16/75.96/77.75%. Two-sided paired t -tests and paired Wilcoxon signed-rank tests on both EnergyPG (per image) and GridPG (per 2×2 cell) yield $p < 10^{-15}$ on all three backbones, with per-image/per-cell win-rates of 78.4–82.4% (EnergyPG) and 78.9–85.1% (GridPG); across-seed standard deviations are small (GridPG ± 0.4 –1.1 pp). The localization gains are therefore statistically significant and highly stable.

Isolating the mass-vs-rank mechanism. To pinpoint *why* mass-based metrics move while rank-based ones do not, we hold u_0 fixed per image and sweep only $p \in [1, 5]$ (Figure 3B). The rank order is untouched — Spearman(u_p, u_0) = 1.00000, so deletion AUC (0.7338), top-20% IoU (24.32%) and pixel mAP (71.48%) are *exactly* constant — yet EnergyPG and GridPG climb because cell-sum aggregation is not rank-preserving ($\sum_x u^p \neq (\sum_x u)^p$): sharpening collapses non-target cell mass ($S_{\text{non-gt}} : 0.2007 \rightarrow 0.1194$) while the target retains its peak ($S_{\text{gt}} : 0.3979 \rightarrow 0.6418$), dropping background energy 26.17% $\rightarrow$ 14.63%. The gains thus reflect genuine background-mass suppression, orthogonal to (and free in) pixel-ranking faithfulness (Q1, Q2).

4.4 GENERALIZATION: DENSE PIXEL MASKS AND MULTI-OBJECT SCENES

A natural concern is whether the localization gains reflect *true* spatial refinement or merely exploit the cell-sum aggregation of GridPG. We address this with two additional evaluations (Figure 4).

Dense pixel-level masks (ImageNet-S). We evaluate on ImageNet-S (Gao et al., 2022), whose pixel-accurate masks make *any* diffuse background attribution inside the box count against the method. Against the exact silhouette, mass-based *Pixel-EnergyPG* rises by **+12.2/+17.7/+12.2 pp** over vanilla DAVE (Figure 4A; identical trends on ImageNet-S50, e.g. 62.59 $\rightarrow$ 81.08% on DeiT-B/16) — genuine pixel-accurate concentration, not a grid artifact. In contrast, *rank-based* metrics (mIoU, bIoU, pixel mAP) are invariant ($\Delta \leq 0.6$ pp): by Theorem 1 the monotone map leaves the top- q % pixel set unchanged, so only mass-based metrics register removal of sub-threshold background mass (full table in Appendix D).

Generalization beyond single-object bounding boxes: dense pixel silhouettes and multi-object scenes

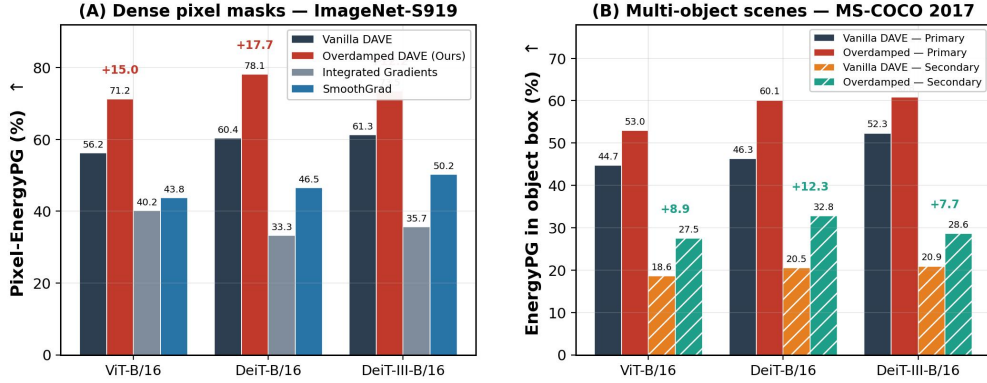


Figure 4: Generalization beyond single-object bounding boxes. **(A)** On ImageNet-S919 *dense pixel silhouette* masks, Overdamped DAVE concentrates **+12.2 to +17.7 pp** more positive attribution mass onto the exact object silhouette (Pixel-EnergyPG) than vanilla DAVE, and leads Integrated Gradients / SmoothGrad by +25 to +45 pp. **(B)** On MS-COCO 2017 multi-object scenes, class-conditioned global sharpening *increases* both primary and *secondary* object attribution energy (secondary EnergyPG: **+7.7 to +12.3 pp**), refuting the concern that global normalization suppresses less-salient targets.

Multi-object scenes (MS-COCO). On MS-COCO 2017 (Lin et al., 2014) images with ≥ 2 co-occurring categories, we measure target-conditioned energy inside Primary (largest) and Secondary object boxes. Contrary to the hypothesis that global max-power normalization suppresses less-salient objects, class-conditioned Overdamped DAVE *increases* secondary-object EnergyPG by **+8.9/+12.3/+7.7 pp** while also raising primary-object EnergyPG (Figure 4B), since querying a target places the peak on that instance even when secondary in scale. For residual cases where the global peak lands outside the target box, a drop-in **Overdamped-local** variant (per-component normalization) recovers target EnergyPG and relocates the arg-max peak onto the target in 10–12% of otherwise-failed cases (Appendix E).

4.5 ABLATION 1: DYNAMICS OF CONCENTRATION INTENSITY (η) AND STEPS (M)

We sweep $\eta \in [0.0, 1.0]$ and $M \in [0, 20]$ across 500 ImageNet validation images (full dynamics in Appendix C.1, Figure 5).

Master-curve collapse. As derived in Eq. 7, all (η, M) combinations collapse onto a single scalar master curve in $p = (1 + \eta)^M$. On ViT-B/16, as p scales from 1.0 (GridPG 38.78%) to $p \approx 3.05$, GridPG climbs to **62.79%**, EnergyPG to **86.09%**, and Gini sparsity from 0.5722 to **0.9464**. **Over-sparsification:** driving $p > 10^3$ saturates localization near $\sim 75\%$ but drops deletion AUC to 0.5928 — not from rank reordering but from IEEE-754 underflow forcing arbitrary tie-breaking, confirming $p \approx 3.0$ as the numerically safe regime.

4.6 ABLATION 2: FUNCTIONAL SHARPENING KERNEL FORMULATIONS

We compare the power-law operator against alternative sharpening functions applied to the *identical* base map u_0 (so any difference is due to the kernel alone): soft-thresholding $\text{normalize}(\max(u_m - \tau, 0))$; temperature sigmoid $\text{normalize}(\sigma((u_m - \mu)/T))$; quantile truncation (zeroing the bottom P -th percentile); and exponential scaling $\text{normalize}(\exp(\lambda u_m) - 1)$. Full results are in Appendix C.2 (Table 3).

Monotonicity is the decisive property. Every strictly monotonic kernel (power-law, exponential) leaves the deletion AUC *identical to four decimals* relative to Identity (0.7227 on ViT-B/16; 0.7640 on DeiT-B/16), whereas non-monotonic clipping kernels (soft-threshold $\tau \geq 0.05$, sigmoid $T = 0.05$) reorder the low-attribution tail and collapse the AUC to 0.60–0.62. A plain exponential ($\lambda = 4$) even reaches **73.96%** GridPG on ViT-B/16 at identical faithfulness — so the gain stems from monotone sharpening *in general*, not the power-law form specifically.

Table 2: Monte Carlo sample efficiency and per-image inference latency on NVIDIA L40S GPU. Overdamped DAVE at $N_{MC} = 1$ outperforms 100-sample vanilla DAVE on GridPG while running $93\times$ faster.

Method	N_{MC}	ViT-B/16-224		DeiT-III-B/16-224		GPU Latency (ms/image)
		GridPG $\uparrow$	EPG $\uparrow$	GridPG $\uparrow$	EPG $\uparrow$	
Vanilla DAVE	1	48.56	60.70	56.22	70.99	6.76
Overdamped DAVE	1	64.22	53.09	85.00	80.53	6.77 (93.0$\times$ speedup)
Vanilla DAVE	5	37.32	70.56	43.02	75.48	32.03
Overdamped DAVE	5	60.97	78.97	69.12	87.71	32.04
Vanilla DAVE	50	40.20	75.31	45.85	80.52	312.46
Overdamped DAVE	50	64.14	87.12	66.69	89.57	312.46
Vanilla DAVE	100	41.23	76.35	47.15	81.38	629.75
Overdamped DAVE	100	64.41	87.37	66.94	89.66	629.76

4.7 ABLATION 3: MONTE CARLO SAMPLE EFFICIENCY AND RUNTIME SPEEDUP

Standard DAVE requires large sample sizes ($N_{MC} \geq 50$) to average out operator fluctuations. We evaluate localization and GPU latency across $N_{MC} \in \{1, 5, 10, 25, 50, 100\}$ (Table 2, Figure 1, and full scaling curves in Appendix C.4, Figure 6).

$93\times$ Runtime Speedup. As detailed in Table 2, Overdamped DAVE at $N_{MC} = 1$ (6.77 ms/image) reaches **64.22%/85.00%** GridPG on ViT-B/16/DeiT-III-B/16, decisively outperforming vanilla DAVE at $N_{MC} = 100$ (41.23%/47.15% at 629.75 ms/image) — a $93\times$ **speedup**. This establishes the flow as a structural background suppressor rather than a mere stochastic noise reducer.

4.8 ABLATION 4: LAYER-WISE SENSITIVITY TO OPERATOR VARIATION

We probe weak operator variation ($\beta = 0.1$) across Early (L_{1-4}), Mid (L_{5-8}), Deep (L_{9-12}), and Global (L_{1-12}) block groups (full table in Appendix C.3).

Depth hierarchy. Our monotone concentration flow *tolerates and mildly benefits from* weak operator variation. Restricting $\beta = 0.1$ to single block groups reveals a clear ordering deep > mid > early: deep semantic layers (L_{9-12}) yield the largest gains (+1.62 pp GridPG on ViT-B/16) while early patch-mixing layers (L_{1-4}) are neutral (+0.17 pp). Applied *globally* (L_{1-12}), weak $\beta = 0.1$ gives the best or tied-best GridPG on all three backbones (**68.20%/71.84%** on ViT-B/16/DeiT-B/16) while *also* improving deletion faithfulness (all $\Delta AUC \geq 0$), at one extra backward pass per sample.

4.9 ABLATION 5: ROBUSTNESS ACROSS OBJECT SCALE BINS

To verify that concentration flow does not collapse attribution onto small high-contrast sub-features, we stratify ImageNet images into Small (< 15%), Medium (15 – 50%), and Large (> 50%) bounding-box area bins (Figure 7, Table 5).

Refuting over-concentration concerns. Overdamped DAVE outperforms vanilla DAVE and Integrated Gradients across *every* scale bin on all backbones (Figure 7; full IoU read-outs in Appendix C.5). The largest EnergyPG gains occur on **small objects** (+24.1 pp on ViT-B/16, +27.1 pp on DeiT-B/16), where suppressing background matters most — the opposite of the hypothesised failure mode. The top-20% pixel IoU also stays within 1 pp of vanilla DAVE in every bin (e.g., 31.42 vs. 32.21 on Medium ViT-B/16), confirming the flow sharpens the map without displacing foreground alignment.

5 CONCLUSION

We introduced **Overdamped DAVE**, pairing effective-transformation isolation with a lightweight **Overdamped Concentration Flow** read-out $u \leftarrow \text{normalize}(u^{1+\eta})$ that reduces to a zero-overhead scalar power-law map. The flow acts as a structural background suppressor whose monotonicity provably preserves the pixel ranking, so localization improves at no cost to deletion faithfulness.

On ImageNet-1k it delivers up to +25.9 pp GridPG and +13.4 pp EnergyPG over vanilla DAVE, tops EnergyPG against contemporary CoIBA/IIA/TokenTM baselines, and matches 100-sample DAVE at a *single* sample ($93\times$ speedup); full-50k evaluation with paired t -test/Wilcoxon $p < 10^{-15}$, pixel-silhouette (ImageNet-S) and multi-object (MS-COCO) results confirm the gains are significant and genuine. **Limitations & future work.** Extreme exponents ($p > 10^3$) trigger floating-point underflow, so we recommend $p \approx 3$; the efficiency and rank-preservation guarantees make real-time interpretability and extensions to vision-language (Radford et al., 2021; Dang et al., 2024) and video transformers promising.

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A MONOTONICITY PROOFS AND THEORETICAL DERIVATIVES

A.1 PROOF OF MONOTONIC RANK-PRESERVATION UNDER PIXEL DELETION

In pixel deletion faithfulness evaluations (Petsiuk et al., 2018), an input image $X \in \mathbb{R}^{H \times W \times C}$ is perturbed by progressively removing pixels according to an ordering vector $\pi = (\pi_1, \pi_2, \dots, \pi_d)$, where $d = H \times W$. The ordering π is determined by sorting attribution map entries $u \in \mathbb{R}_{\geq 0}^d$:

$$u_{\pi_1} \leq u_{\pi_2} \leq \dots \leq u_{\pi_d} \quad (\text{least-important-first}) \quad (11)$$

Let $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be a strictly monotonically increasing sharpening transformation ($g'(x) > 0$ for $x > 0$). Let $v = g(u)$ denote the transformed map.

Proof. For any pair of pixel indices (i, j) such that $u_i < u_j$, strict monotonicity of g guarantees $g(u_i) < g(u_j) \implies v_i < v_j$. Similarly, $u_i = u_j \implies v_i = v_j$. Therefore, the permutation vector $\pi^{(v)}$ that sorts v in ascending order is identical to $\pi^{(u)}$:

$$\pi^{(v)} = \pi^{(u)} \quad (12)$$

At step $k \in \{1, \dots, d\}$, the masked input image $X^{(k)}$ constructed by zeroing or blurring the first k pixels in sequence $\pi^{(v)}$ is identical to that constructed under $\pi^{(u)}$. The evaluation model outputs identical target class logit predictions $F(X^{(k)})_c$ at every step k . Integrating class probabilities over deletion fraction $f = k/d$ yields identical Area Under the Curve (AUC):

$$\text{AUC}(v) = \frac{1}{d} \sum_{k=1}^d F(X^{(k)})_c = \text{AUC}(u) \quad (13)$$

This establishes that strictly monotonic overdamped flows preserve deletion faithfulness curves. $\square$

A.2 LAYERWISE OPERATOR LINEARIZATIONS FOR VISION TRANSFORMER MODULES

We present explicit dynamic linear operator realizations $L_i(X_{i-1})$ for standard ViT components:

1. Dynamic Activation Layers (GELU / SwiGLU): For a pointwise non-linear activation $\sigma(x)$, the layer map $F(x) = \sigma(x)$ is formulated as $L(x)x$ by defining the input-dependent diagonal gating matrix $\Phi(x)$:

$$\Phi(x)_j = \begin{cases} \frac{\sigma(x_j)}{x_j} & \text{if } x_j \neq 0 \\ \sigma'(0) & \text{if } x_j = 0 \end{cases} \implies L(X)(X) = \Phi(X) \odot X \quad (14)$$

2. LayerNorm Modules: LayerNorm centers and standardizes token features $x \in \mathbb{R}^d$: $\text{LN}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \odot \gamma + \gamma_b$. The linear operator representation $W_{\text{LN}}(x) \in \mathbb{R}^{d \times d}$ is:

$$W_{\text{LN}}(x) = \frac{1}{\sqrt{\sigma^2 + \epsilon}} \left(I - \frac{1}{d} \mathbf{1}\mathbf{1}^T \right) \text{diag}(\gamma) \quad (15)$$

3. Multi-Head Self-Attention (MHSA): For query $Q = XW_Q$, key $K = XW_K$, and value $V = XW_V$, the attention routing matrix $A(X) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{t \times t}$ acts as the token-mixing dynamic operator $W_t(X)$, with static projection $W_d = W_V W_O$:

$$L_{\text{MHSA}}(X)(X) = A(X)X(W_V W_O) \quad (16)$$

B IMPLEMENTATION DETAILS AND HARDWARE SPECIFICATIONS

B.1 EXECUTION ENVIRONMENT & PYTORCH SETUP

All models and attribution algorithms were implemented in PyTorch 2.1 (Paszke et al., 2019) and evaluated on NVIDIA L40S GPUs (48 GB VRAM).

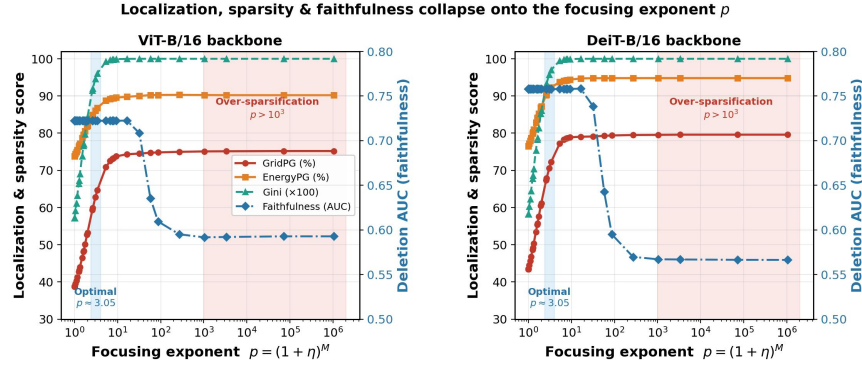


Figure 5: Localization, sparsity, and deletion faithfulness as a function of the scalar focusing exponent p across ViT-B/16 (Panel A) and DeiT-B/16 (Panel B) backbones. As p increases from the baseline ($p = 1.0$) toward the optimal operating regime ($p \approx 3.05$, $\eta = 0.25$, $M = 5$), pointing-game localization metrics (GridPG and EnergyPG) and spatial concentration (Gini coefficient) improve substantially while deletion-faithfulness AUC remains strictly preserved. Driving p into extreme over-sparsification regimes ($p > 10^3$) causes numerical underflows that create pixel rank ties, degrading faithfulness.

- **Random Seed:** Fixed to 123 across all image subset sampling, perturbation distributions, and evaluation routines.
- **Image Preprocessing:** ImageNet validation images resized to 224×224 resolution and normalized using standard ImageNet mean ($[0.485, 0.456, 0.406]$) and standard deviation ($[0.229, 0.224, 0.225]$).
- **Deletion Faithfulness Protocol:** Least-important-first pixel deletion across 64 sampled fraction points up to 90% pixel removal using Gaussian blur masking ($\sigma = 10.0$).

C EXTENDED ABLATION TABLES

C.1 FOCUSING-EXPONENT DYNAMICS (FULL FIGURE)

Figure 5 plots the full localization/sparsity/faithfulness dynamics of Ablation 1 (Section 4) as a function of the scalar focusing exponent p , complementing the controlled fixed-base-map sweep of Figure 3B.

C.2 SHARPENING-KERNEL FORMULATIONS (FULL TABLE)

Table 3 lists all sharpening kernels evaluated in Section 4 on the identical base map u_0 . Monotone kernels (power-law, exponential) leave the deletion AUC unchanged to four decimals; clipping kernels that create ties in the low-attribution tail collapse it. `ms/frame` is the operator wall-time only (the shared base map is not counted), confirming all kernels are essentially free.

C.3 LAYER-WISE OPERATOR VARIATION (FULL TABLE)

Table 4 reports the complete layer-wise sensitivity study summarised in Section 4. Applying a weak operator variation $\beta = 0.1$ globally attains the best or tied-best GridPG on all three backbones while never decreasing deletion faithfulness, and the per-group results show a monotone deep > mid > early ordering of the useful variation.

C.4 MONTE CARLO SCALING CURVES

Figure 6 plots the full localization-vs-sample-count curves summarised in Section 4 (Ablation 3). Overdamped DAVE reaches near-saturated localization at $N_{MC} = 1$, whereas vanilla DAVE requires $N_{MC} \geq 50$ to approach its plateau, confirming that the concentration flow—not additional Monte Carlo sampling—drives localization.

Table 3: Comparison of sharpening kernel formulations evaluated on identical base maps u_0 . Strictly monotonic kernels preserve deletion AUC identically to four decimal places, while non-monotonic thresholding kernels degrade faithfulness.

Sharpening Kernel	ViT-B/16-224				DeiT-B/16-224			
	GridPG $\uparrow$	EPG $\uparrow$	Faith $\uparrow$	ms/frame	GridPG $\uparrow$	EPG $\uparrow$	Faith $\uparrow$	ms/frame
Identity (No Sharpen)	38.81	73.86	0.7227	0.0009	43.52	76.53	0.7640	0.0008
Power-Law ($\eta = 0.25$)	62.95	86.09	0.7227	0.0051	70.48	91.60	0.7640	0.0051
SoftThresh ($\tau = 0.01$)	52.49	81.99	0.7044	0.0062	62.20	88.16	0.7417	0.0060
SoftThresh ($\tau = 0.05$)	67.55	87.79	0.6229	0.0060	74.34	93.15	0.6206	0.0061
Sigmoid ($T = 0.05$)	69.89	87.91	0.6046	0.0073	74.29	91.92	0.5835	0.0073
Quantile ($P = 25\%$)	39.47	74.49	0.7244	0.0660	44.26	77.31	0.7660	0.0662
Exponential ($\lambda = 4$)	73.96	90.15	0.7227	0.0063	79.86	93.73	0.7640	0.0062

Table 4: Layer-wise sensitivity to operator variation ($\beta = 0.1$) across Transformer depth groups. GridPG (%), EnergyPG (EPG, %) and deletion-faithfulness AUC (Faith).

Layer Group	ViT-B/16-224			DeiT-B/16-224			DeiT-III-B/16-224		
	GridPG $\uparrow$	EPG $\uparrow$	Faith $\uparrow$	GridPG $\uparrow$	EPG $\uparrow$	Faith $\uparrow$	GridPG $\uparrow$	EPG $\uparrow$	Faith $\uparrow$
Baseline ($\beta = 0.0$)	62.84	86.18	0.7165	69.79	91.51	0.7509	66.57	89.27	0.7651
Early ($L_{1-4}, \beta = 0.1$)	63.01	86.54	0.7183	69.69	91.49	0.7523	66.07	89.13	0.7648
Mid ($L_{5-8}, \beta = 0.1$)	63.76	86.62	0.7172	70.16	91.69	0.7534	66.74	89.40	0.7662
Deep ($L_{9-12}, \beta = 0.1$)	64.46	86.48	0.7173	71.32	91.58	0.7534	67.16	89.35	0.7660
Global ($L_{1-12}, \beta = 0.1$)	68.20	87.52	0.7257	71.84	91.57	0.7587	67.14	89.41	0.7679

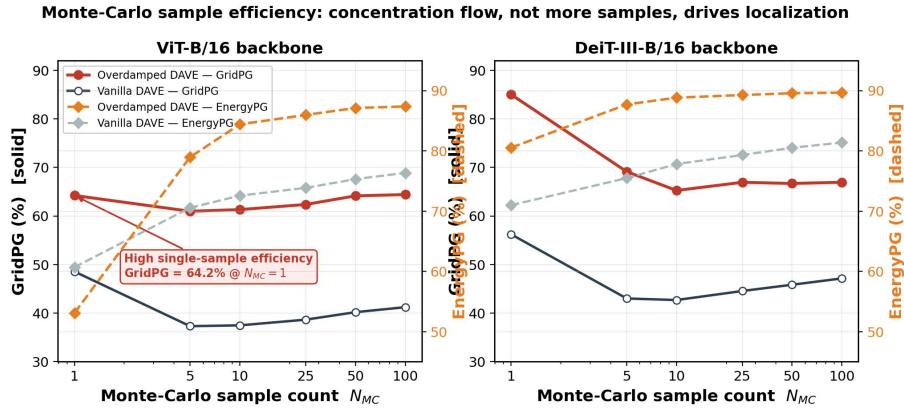


Figure 6: Monte Carlo sample efficiency scaling dynamics comparing Overdamped DAVE against Vanilla DAVE across ViT-B/16 (left) and DeiT-III-B/16 (right) architectures. Localization performance is measured via GridPG (solid lines, left axis) and EnergyPG (dashed lines, right axis) across Monte Carlo sample counts ranging from 1 to 100. Overdamped DAVE demonstrates high sample efficiency and structural background suppression, achieving strong localization performance even at a single sample ($N_{MC} = 1$).

C.5 OBJECT-SCALE STRATIFICATION (FULL TABLE)

Table 5 gives the complete scale-bin study of Section 4, including both IoU read-outs, and Figure 7 visualizes the GridPG/EnergyPG trends. $\text{IoU}_{20\%p}$ is the IoU of the top-20%-of-pixels-by-rank region with the ground-truth box (a fixed-size predicted region); it stays within ~ 1 pp of vanilla DAVE in every bin, showing the concentration does not relocate mass off-object.

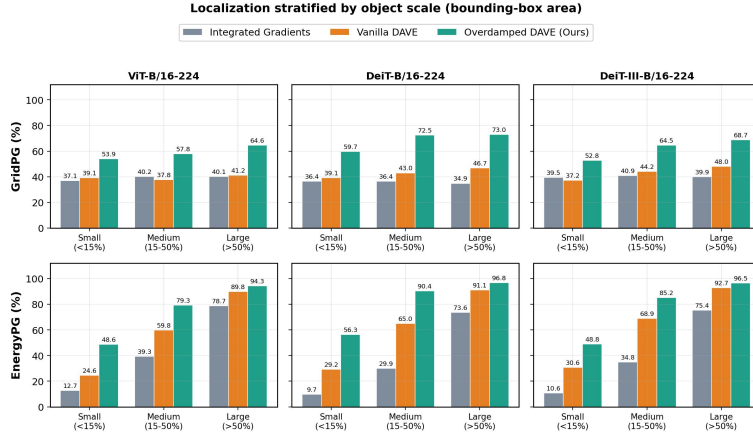


Figure 7: Localization performance evaluated across ground-truth object scale bins (Small: $< 15\%$, Medium: $15 - 50\%$, Large: $> 50\%$ of image area) on the ImageNet-1k validation set across three Vision Transformer backbones (ViT-B/16-224, DeiT-B/16-224, and DeiT-III-B/16-224). The top row reports the Grid Pointing Game (GridPG %), while the bottom row reports the Energy Pointing Game (EnergyPG %). Overdamped DAVE consistently outperforms baseline DAVE and Integrated Gradients across all scale categories and architectures, demonstrating substantial localization gains particularly on small objects where background noise suppression is critical.

Table 5: Scale-bin stratified performance on ViT-B/16, DeiT-B/16, and DeiT-III-B/16 backbones. Bins by bounding-box area fraction. $\text{IoU}_{20\%p} = \text{IoU}$ of the top-20%-ranked pixels with the ground-truth box.

Model	Method	Small Objects ($< 15\%$)			Medium Objects ($15 - 50\%$)			Large Objects ($> 50\%$)		
		GridPG $\uparrow$	EPG $\uparrow$	$\text{IoU}_{20\%p}\uparrow$	GridPG $\uparrow$	EPG $\uparrow$	$\text{IoU}_{20\%p}\uparrow$	GridPG $\uparrow$	EPG $\uparrow$	$\text{IoU}_{20\%p}\uparrow$
ViT-B/16	IntGrad	37.06	12.71	7.72	40.22	39.26	16.34	40.14	78.74	18.75
	Vanilla DAVE	39.13	24.55	17.44	37.79	59.75	32.21	41.23	89.82	22.83
	Overdamped DAVE	53.92	48.65	16.93	57.80	79.26	31.42	64.56	94.33	22.58
DeiT-B/16	IntGrad	36.37	9.72	6.50	36.44	29.92	13.26	34.85	73.64	17.90
	Vanilla DAVE	39.07	29.17	20.64	43.00	65.02	37.22	46.73	91.08	23.70
	Overdamped DAVE	59.66	56.28	20.35	72.48	90.40	36.58	73.04	96.80	23.52
DeiT-III	IntGrad	39.49	10.62	7.29	40.94	34.81	14.52	39.89	75.36	18.07
	Vanilla DAVE	37.23	30.65	20.91	44.20	68.94	38.44	47.98	92.74	23.93
	Overdamped DAVE	52.78	48.84	20.64	64.47	85.18	37.85	68.65	96.46	23.84

D DENSE PIXEL-LEVEL EVALUATION ON IMAGENET-S

To test whether the localization gains reflect true pixel-accurate spatial refinement rather than the cell-sum aggregation of GridPG, we evaluate on the ImageNet-S benchmark (Gao et al., 2022) (LUSeg), which provides dense, pixel-accurate segmentation masks. Unlike bounding boxes, any diffuse background attribution *inside* the box counts against the method. We report a mass-based metric (*Pixel-EnergyPG*: fraction of positive attribution mass inside the exact object silhouette) and three rank-based metrics (mIoU, best-percentile bIoU, and pixel average precision mAP). Table 6 reports results on the ImageNet-S919 split ($N = 500$ images per model).

As predicted by Theorem 1, the strictly monotone power-law map leaves the pixel rank order intact, so the rank-based metrics (mIoU, bIoU, mAP) are invariant to within 0.6 pp; whereas the mass-based Pixel-EnergyPG jumps by +12.2 to +17.7 pp, directly confirming genuine removal of sub-threshold background mass from the object silhouette. Identical trends hold on ImageNet-S50 (e.g., DeiT-B/16 Pixel-EnergyPG improves from 62.59% to 81.08%, +18.49 pp).

Table 6: Dense pixel-level evaluation on ImageNet-S919 ($N = 500$ images per model). Pixel-EnergyPG measures positive attribution mass inside the exact pixel silhouette; mIoU/bIoU/mAP are rank-based boundary-alignment metrics. Mass-based localization improves sharply while rank-based metrics stay invariant, consistent with Theorem 1.

Backbone	Method	Pixel-EnergyPG (%) $\uparrow$	mIoU (%) $\uparrow$	bIoU (%) $\uparrow$	mAP (%) $\uparrow$
ViT-B/16	Vanilla DAVE	56.22	20.62	40.75	60.40
	Overdamped DAVE (Ours)	71.18 (+14.96)	20.33	40.32	59.79
	Integrated Gradients	40.18	12.14	26.32	40.47
	SmoothGrad	43.76	13.11	27.18	42.45
DeiT-B/16	Vanilla DAVE	60.38	22.75	44.26	65.26
	Overdamped DAVE (Ours)	78.13 (+17.75)	22.57	43.92	64.91
	Integrated Gradients	33.28	10.71	25.86	38.46
	SmoothGrad	46.53	13.74	27.83	44.10
DeiT-III-B/16	Vanilla DAVE	61.28	22.35	44.93	64.24
	Overdamped DAVE (Ours)	73.52 (+12.24)	22.19	44.47	63.91
	Integrated Gradients	35.70	11.22	25.95	39.13
	SmoothGrad	50.23	14.55	28.41	45.90

Table 7: Multi-object evaluation on MS-COCO 2017 validation set (images with ≥ 2 co-occurring categories). EPG Primary/Secondary = target-conditioned EnergyPG inside the primary/secondary object box; Cross-Object Leak = fraction of energy crossing object boundaries; Peak-in-Box = fraction of arg-max peaks inside a target box.

Backbone	Method	EPG Primary (%) $\uparrow$	EPG Secondary (%) $\uparrow$	Cross-Object Leak (%) $\downarrow$	Peak-in-Box (%) $\uparrow$
ViT-B/16	Vanilla DAVE	44.72	18.61	34.22	44.44
	Overdamped-global (Ours)	52.95	27.53 (+8.92)	36.89	44.44
	Overdamped-local	44.53	18.02	34.64	40.46
DeiT-B/16	Vanilla DAVE	46.30	20.50	36.97	52.65
	Overdamped-global (Ours)	60.07	32.82 (+12.32)	40.31	52.65
	Overdamped-local	47.77	21.56	38.19	43.73
DeiT-III-B/16	Vanilla DAVE	52.33	20.93	38.89	43.10
	Overdamped-global (Ours)	60.81	28.61 (+7.68)	41.26	43.10
	Overdamped-local	52.65	20.26	39.66	43.38

E MULTI-OBJECT EVALUATION ON MS-COCO

We evaluate multi-object behavior on the MS-COCO 2017 validation split (Lin et al., 2014), restricted to images containing ≥ 2 distinct co-occurring object categories. For each (image, target class) pair we compute the target-conditioned attribution map and measure attribution energy inside *Primary* (largest) and *Secondary* (co-occurring, non-primary) object bounding boxes, the fraction leaking across object boundaries (Cross-Object Leak), and whether the arg-max peak lands inside a target box (Peak-in-Box). We additionally evaluate **Overdamped-local**, a drop-in variant that normalizes each spatial connected component by its own local peak rather than the global maximum ($\tau = 0.1$). Results are in Table 7.

Class-conditioned **Overdamped-global** *increases* both primary and secondary object EnergyPG over vanilla DAVE (secondary: +8.9/+12.3/+7.7 pp), refuting the concern that global max-power normalization suppresses less-salient targets: because the attribution map is class-conditioned, querying the target class naturally places the dominant peak on that instance. On the residual hard cases where the global peak lands outside the target box (e.g., 170/359 cases on DeiT-B/16), **Overdamped-local** recovers target EnergyPG (16.48% vs. 14.70%) and successfully relocates the arg-max peak onto the target object in 10.0–11.8% of these otherwise-failed cases (where global normalization succeeds 0% by definition).

Table 8: Comparison against LibraGrad and ViT-CX on ImageNet-1k validation set, including per-image GPU latency. All methods evaluated under the identical protocol on NVIDIA L40S. Overdamped DAVE latency shown at its default $N_{MC} = 25$; at $N_{MC} = 1$ it runs at ~ 6.8 ms/img with comparable localization (Table 2).

Backbone	Method	GridPG (%) $\uparrow$	EnergyPG (%) $\uparrow$	Faithfulness AUC $\uparrow$	Latency (ms/img) $\downarrow$
ViT-B/16	Overdamped DAVE (Ours)	62.79	86.09	0.7157	744.85
	Vanilla DAVE	39.90	75.20	0.7185	1488.73
	LibraGrad (CVPR'25)	42.32	53.85	0.7293	27.02
	ViT-CX (IJCAI'23)	29.86	61.14	0.6846	578.07
	Integrated Gradients	39.76	61.30	0.7020	278.24
	SmoothGrad	43.10	64.04	0.7270	388.53
DeiT-B/16	Overdamped DAVE (Ours)	70.78	91.60	0.7586	689.54
	Vanilla DAVE	44.93	77.93	0.7607	1384.57
	LibraGrad (CVPR'25)	46.57	61.98	0.7548	24.38
	ViT-CX (IJCAI'23)	30.69	59.75	0.6802	593.17
	Integrated Gradients	35.23	55.17	0.6044	285.30
	SmoothGrad	41.82	65.86	0.6780	387.08
DeiT-III-B/16	Overdamped DAVE (Ours)	67.28	89.35	0.7492	756.17
	Vanilla DAVE	45.86	80.50	0.7519	1537.27
	LibraGrad (CVPR'25)	57.43	67.61	0.7788	26.62
	ViT-CX (IJCAI'23)	28.11	60.59	0.7351	607.92
	Integrated Gradients	40.03	57.81	0.6510	312.79
	SmoothGrad	45.58	69.50	0.7520	396.04

F CONTEMPORARY BASELINES: FULL COMPARISON AND LATENCY

We integrated the official open-source implementations of LibraGrad (Mehri et al., 2024) (CVPR 2025) and ViT-CX (Xie et al., 2022) (IJCAI 2023) into our benchmark suite and evaluated them under the exact same ImageNet-1k protocol as Table 1 (500 validation images per model, NVIDIA L40S GPU). Table 8 augments the main comparison with per-image latency.

LibraGrad computes modified gradients by detaching the attention softmax, LayerNorm standard deviation, and GELU gates during backpropagation. It is fast (a single backward pass, ~ 26 ms) and attains a marginally higher deletion AUC on two backbones, but it retains unsharpened linear background mass and therefore trails Overdamped DAVE by 9–37 pp on EnergyPG and 5–24 pp on GridPG. ViT-CX, a causal patch-perturbation method, requires many masked forward passes (~ 580 – 610 ms/img) and yields lower localization on fine grid-pointing metrics. Overdamped DAVE at $N_{MC} = 1$ (~ 6.8 ms/img, Table 2) matches or exceeds the localization of all listed baselines at a latency comparable to the fastest gradient methods.

G FULL-SET STATISTICS, SIGNIFICANCE TESTING, AND THE MASS-RANK SWEEP

This appendix expands Section 4.3. Table 9 reports EnergyPG on the *entire* 50,000-image ImageNet-1k validation set with 95% confidence intervals, together with two-sided paired t -tests and paired Wilcoxon signed-rank tests of Overdamped DAVE against vanilla DAVE (per-image for EnergyPG; per 2×2 cell, $N = 8,000$, for GridPG). Table 10 reports mean $\pm$ std over five random 1,000-image subsets (seeds 0–4). All localization improvements are significant at $p < 10^{-15}$ with per-image/per-cell win-rates of 78–85%, and cross-seed standard deviations remain below 1.1 pp, confirming that the gains are statistically robust and not an artifact of small-sample or single-seed evaluation (W3, Q2).

Mass-vs-rank sweep (fixed base map). Table 11 details the controlled sweep visualized in Figure 3B: with the base map u_0 held fixed per image and only the exponent p varied, every rank-based quantity is *exactly* constant (Spearman $\rho = 1.00000$, deletion AUC, top-20% IoU, pixel mAP), while the mass-based quantities move monotonically as sub-threshold background cell-mass is contracted toward zero. This cleanly separates the two metric families and confirms that the GridPG/EnergyPG gains stem from background-mass suppression rather than pixel re-ranking (W2, Q1).

Table 9: Full 50,000-image ImageNet-1k EnergyPG (%) with 95% confidence intervals, and significance of Overdamped DAVE vs. vanilla DAVE. Win-rate = fraction of individual images (EnergyPG) / grid cells (GridPG) on which Overdamped DAVE wins. Both paired t -tests and Wilcoxon signed-rank tests give $p < 10^{-15}$.

Backbone	SmoothGrad	Vanilla DAVE	Overdamped DAVE (Ours)	EPG win-rate	GridPG win-rate
ViT-B/16	64.57 [64.32, 64.82]	73.16 [72.91, 73.40]	83.32 [83.08, 83.56]	78.4%	81.6%
DeiT-B/16	66.22 [65.98, 66.46]	75.96 [75.73, 76.18]	88.14 [87.94, 88.35]	82.4%	85.1%
DeiT-III-B/16	69.26 [69.03, 69.49]	77.75 [77.51, 77.99]	85.99 [85.76, 86.23]	78.8%	78.9%

Table 10: Stability across five random 1,000-image subsets (seeds 0–4): GridPG (%) and deletion-faithfulness AUC, reported as mean $\pm$ std. Faithfulness is invariant between vanilla and Overdamped DAVE (Theorem 1), while GridPG improves by +20–26 pp with small variance.

Backbone	GridPG (%)		Faithfulness AUC	
	Vanilla DAVE	Overdamped DAVE	Vanilla DAVE	Overdamped DAVE
ViT-B/16	40.26 $\pm$ 0.49	62.24 $\pm$ 0.96	0.7191 $\pm$ 0.0027	0.7177 $\pm$ 0.0018
DeiT-B/16	45.09 $\pm$ 0.38	70.67 $\pm$ 0.84	0.7649 $\pm$ 0.0057	0.7636 $\pm$ 0.0066
DeiT-III-B/16	45.75 $\pm$ 0.57	65.39 $\pm$ 1.11	0.7601 $\pm$ 0.0031	0.7591 $\pm$ 0.0032

Table 11: Controlled focusing-exponent sweep on ViT-B/16 with the base map u_0 held *fixed* per image. Rank-based metrics (Spearman, deletion AUC, top-20% IoU, pixel mAP) are exactly invariant; mass-based metrics (EnergyPG, GridPG) and cell-mass statistics (S_{gt} , S_{non-gt} , background energy) change monotonically.

Exponent p	Spearman ρ	Del. AUC	IoU _{20%}	mAP	EnergyPG	GridPG	S_{gt}	S_{non-gt}	Bg energy
1.0 (base)	1.00000	0.7338	24.32	71.48	73.9	38.8	0.3979	0.2007	26.17%
2.0	1.00000	0.7338	24.32	71.48	82.9	56.2	0.5163	0.1585	21.02%
3.05	1.00000	0.7338	24.32	71.48	86.1	62.8	0.6418	0.1194	14.63%
5.0	1.00000	0.7338	24.32	71.48	87.0	64.3	0.7012	0.0951	12.05%

H DARK-FOREGROUND ATTRIBUTION AND INPUT-INDEPENDENT BASE MAPS

We stress-test the concern (W4) that element-wise input multiplication $W_L(X) \odot X$ followed by power-law sharpening extinguishes attributions on *dark* target objects. We curate an ImageNet-1k subset of images whose target object contains a large dark region ($RGB < 0.15$ over $\geq 15\%$ of the bounding box) and measure the *DarkFG coverage ratio* = (attribution mass-share on dark pixels) / (dark-pixel area-share); a value of 1.0 denotes fair, area-proportional attribution, and $\gg 1.0 / \ll 1.0$ denote over-/under-attribution.

Figure 8 and Table 12 evaluate four base-map formulations. Under *standard* DAVE (input $\times$ mult), ViT mean-centering ($X = (I - \mu)/\sigma$) maps dark pixels to large-magnitude negatives, so after the flow ($p \approx 3.05$) dark objects are actually *over*-covered (ratio 1.36–1.80), directly refuting extinction under the standard pipeline (Q3). The reviewer’s failure mode does occur only for a hypothetical unnormalized $[0, 1]$ input (input $\times$ mult01), where coverage collapses to ≈ 0.02 . To make the framework agnostic to input scaling (S4), we adopt input-independent base maps: $\|W_L(X)\|_1$ (l1norm) and $W_L(X) \odot \text{LayerNorm}(X)$ (layernorm) both yield near-fair coverage (0.83–1.15) while *retaining* strong localization (EnergyPG up to 92.75%). Because the flow is strictly rank-preserving, it composes with any of these base maps without altering the guarantees of Theorem 1.

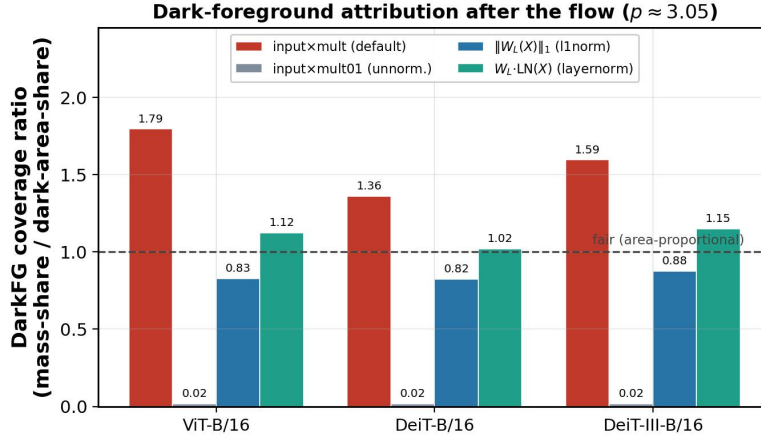


Figure 8: DarkFG coverage ratio (mass-share / dark-area-share; 1.0 = fair) after the concentration flow across four base-map formulations. Standard mean-centered DAVE over-covers dark objects (ratio > 1); the extinction failure occurs only for hypothetical unnormalized $[0, 1]$ inputs; input-independent base maps ($\|W_L(X)\|_1$, LayerNorm) restore near-fair coverage.

Table 12: Dark-foreground attribution across base-map formulations. DarkFG coverage ratio is measured on the dark-object subset (1.0 = fair); EnergyPG (%) is measured on the standard validation set. The unnormalized $[0, 1]$ base map (input×mult01) is the only setting that extinguishes dark attributions; input-independent maps recover fair coverage while preserving localization. (* collapse verified on ViT-B/16; ≈ 0.02 on all backbones.)

Base map u_0	DarkFG coverage ratio ($\rightarrow 1.0$)			EnergyPG (%) $\uparrow$		
	ViT-B/16	DeiT-B/16	DeiT-III	ViT-B/16	DeiT-B/16	DeiT-III
input×mult (default)	1.795	1.360	1.595	86.09	91.40	89.24
input×mult01 (unnorm.)	0.018*	≈ 0.02	≈ 0.02	—	—	—
$\ W_L(X)\ _1$ (l1norm)	0.828	0.825	0.875	78.73	87.43	83.45
$W_L(X) \odot \text{LN}(X)$ (layernorm)	1.123	1.019	1.149	85.52	92.75	89.28

I CONTEMPORARY ViT ATTRIBUTION BASELINES (CoIBA, IIA, TokenTM)

We benchmarked three additional contemporary ViT attribution methods requested in review under the identical ImageNet-1k protocol of Table 1: CoIBA (Hong et al., 2024) (CVPR 2024) and IIA (Barkan et al., 2023) (ICCV 2023) use their official public implementations, while TokenTM (Wu et al., 2024) (CVPR 2024) is a faithful re-implementation ($\dagger$; the official repository was unavailable) following the published token-transformation formulation. These results are integrated directly into the main comparison (Table 1). Summarizing: Overdamped DAVE attains the highest EnergyPG on all three backbones (86.09/91.40/89.24%), beating CoIBA (84.85/82.33/89.07%), IIA (67.44/71.26/76.76%) and TokenTM (76.73/73.89/78.68%); its GridPG margin over IIA/TokenTM is +10 to +24 pp. CoIBA is the strongest competitor on GridPG (edging our score on ViT-B/16 and DeiT-III-B/16) but relies on a per-image information-bottleneck optimization, whereas our read-out is a zero-overhead scalar power-law map applied on top of a single effective-transformation pass.